

Platen Superheater Transient Simulation — Complete User Guide

Application: Unit №4 Platen Superheater Transient Simulator (Android)

Platform: Android 8.0+ (API 26+)

Property engine: IAPWS-IF97 (Regions 1, 2, 4) — vendored `com.hummeling.if97` v2.1.0

Version: 1.0.0

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1. What the application does

This is an **engineering simulation tool**, not a static calculator. It computes how the

outlet steam temperature of the platen superheater in a 325 MW natural-circulation

drum boiler (Unit №4) responds **over time** to:

#	Driver
1	Steam mass flow / load changes
2	Spray-water flow changes at the platen inlet
3	Burner start/stop events
4	Furnace heat-input changes
5	Steam inlet pressure changes
6	Steam inlet temperature changes
7	Property changes with P and T (density ρ , c_p , μ , k)
8	Internal convection coefficient h_i changing with flow and properties
9	Reynolds number changes with load
10	Prandtl number changes with steam state
11	Nusselt number changes with Re and Pr
12	Dynamic overall heat-transfer coefficient U
13	Tube-metal thermal inertia (the dominant delay source)
14	Steam thermal capacity / residence time
15	Spray injection cooling dynamics

The causal chain is physical, top to bottom:

P, T $\rightarrow$ IF97 properties (ρ , μ , c_p , k)
 $\rightarrow$ velocity $\rightarrow$ Re $\rightarrow$ Pr $\rightarrow$ Nu $\rightarrow$ h_i $\rightarrow$ U
 $\rightarrow$ Q (uniform coil)
 $\rightarrow$ metal energy balance $\rightarrow$ steam energy balance
 $\rightarrow$ steam outlet T (with spray enthalpy + burner heat entering dynamically)

U is a calculated, load-dependent output — never a hard-coded constant.

Plant reference data (locked, Unit №4)

Parameter	Value
Boiler capacity	325 MW
Type	Natural circulation, drum-type
Burners	24 × 40 MW, 3 elevations
Full-load steam flow	1000–1040 t/h
Sliding pressure range	10–167 bar
Sliding steam temperature	~200–540 °C
Parallel panels	43
Tubes per panel	4 (172 total)
Passes per panel	8, pass length 6 m
Tube OD / wall / ID	57 / 8 / 41 mm
Header feed	Dual mid-point (panels 1–21 / 22–43)

Uniform platen material (refactored model — the coil is treated as one uniform tube of a single material):

Material	ρ (kg/m ³)	c_p (J/kg·K)	k (W/m·K)
12Cr2MoWVTiB	7850	520	32

Initial platen metal temperature is a user input (default **450 °C**) and enters the calculation as the starting point of the metal energy balance.

2. Installing the app

1. Obtain `app-debug.apk` (built from `./gradlew :app:assembleDebug`, located in `app/build/outputs/apk/debug/`).
1. Copy it to your Android device (USB, cloud, or direct download).
2. On the device, open the APK. Android will ask to allow installation from that source ("Install unknown apps") — allow it once.
1. Tap **Install**, then open **Platen Superheater Transient**.

No special permissions are required. The app works fully offline — the IF97 property library is compiled into the app.

3. How the simulation works — the big picture

The platen coil is modeled as **one uniform tube** (refactored model — segmentation removed for simplicity): a single material (12Cr2MoWVTiB), uniform heat flux, and two lumped state variables:

- **Metal temperature** T_m — the tube-wall metal lump (dominant inertia; starts at the user's Metal T input, default 450 °C)
- **Steam temperature** T_s — the steam in the coil

Each timestep the model:

1. Reads the current inputs (steam flow, spray flow/temperature, burners, pressure, steam temperature) — including any enabled scenario events due at this time.

1. If spray is active, performs the **mixing calculation** (see §5): enthalpy balance → mixed enthalpy → mixed temperature via the IF97 backward equation $T(P,h)$.

1. Evaluates properties (ρ , μ , c_p , k) from IF97 at the current steam state, computes velocity → Re → Pr → Nu → h_i , then U from the cylindrical resistance chain.

1. Computes the heat flows and integrates both energy balances with the **RK4** method (4th-order Runge–Kutta).

1. Records every thermodynamic parameter for the results and charts.

Why the metal matters: the tube metal holds ~36 MJ/K — thousands of times more thermal capacity than the steam in the tubes. When burners step up, the metal heats first and the steam follows; that lag is exactly what the model reproduces.

Flow distribution assumption (explicit): total platen steam flow splits equally across all 43 panels; within a panel the 4 tubes share equally. So one tube carries $\dot{m}_{total} / 172$. At full load (277.8 kg/s) and 167 bar / 520 °C ($\rho \approx 52 \text{ kg/m}^3$) the 41 mm ID gives $\approx 24 \text{ m/s}$ tube velocity and $Re \approx 1.5 \cdot 10^6$ — physically correct for superheated steam.

4. The Simulate tab — step by step

4.1 Initial state fields

Field	Meaning	Default
Steam flow t/h	Total platen steam flow at $t = 0$	300
Pressure bar	Steam pressure at the platen inlet (absolute)	100
Steam T °C	Steam temperature entering the platen at $t = 0$	400
Spray T °C (100–180)	Spray-water temperature — inside the enforced window	150
Metal T °C	Initial platen metal temperature — a user input	450
Burners on	Number of burners firing at $t = 0$	2
Duration min	Simulation length	15

4.2 Scenario cards — enable any combination

Each scenario (steam flow, spray flow, burners, pressure, steam temperature) is a card

with its own **switch**, event **time** and target **value**:

- Toggle the switch **on** to include that scenario in the run.
- Leave it **off** and that scenario never fires — the quantity stays at its initial

value for the whole simulation.

- With one switch on you get a **single-scenario** run; with several on you get a

mixed-scenario run — all enabled scenarios apply at their own times.

4.3 Running

Tap **RUN SIMULATION**. The engine integrates with $dt = 1$ s. Results appear below:

summary numbers first, then the parameter-selection chips and charts, then the

calculation-details panel.

If anything is wrong with the inputs (e.g. spray outside 100–180 °C, non-physical

pressure), a red error card appears with an engineering-readable explanation instead

of a crash.

5. Spray water rules (important)

The app **enforces the plant rule** for spray water:

> **Spray water is always subcooled and between 100 and 180 °C — in any situation.**

Concretely, the engine checks, before every mixing calculation:

1. **Window:** spray temperature must be **between 100 °C and 180 °C**.
2. **Subcooling:** spray temperature must also be ****below the saturation temperature**

Tsat** at the mixing pressure. At 20 bar, Tsat ≈ 212 °C, so the window binds; below
≈ 13 bar, Tsat drops inside the window and subcooling binds first.

If either check fails, the simulation refuses with a typed error explaining exactly
which condition failed and by how much — for example:

```
Spray temperature 200.0 °C outside the 100-180 °C plant window  
Spray water must be subcooled: T=120.0 °C >= Tsat=110.8 °C at 1.50 bar
```

Why: desuperheating sprays must evaporate completely in the steam stream. Water
that is saturated or boiling at the injection pressure flashes rather than mixing,
which the single-zone mixing model does not represent. The rule keeps every result in
the physically valid region.

Spray temperature as an event: you can also change spray temperature during a
scenario (`SPRAY_TEMP_K` events); the same rule is enforced at every event.

The mixing calculation (what happens when spray is on)

Four distinct states are always kept separate:

1. **Upstream steam** (pre-spray): P, T, h from IF97
2. **Spray water:** h from IF97 Region 1 at the injection pressure and spray temperature
3. **Mixed state** (post-spray, platen inlet): enthalpy balance → T(P,h)
4. **Platen outlet:** computed by the uniform-coil energy balance

```
h_mix = (ṁ_steam·h_steam + ṁ_spray·h_spray) / (ṁ_steam + ṁ_spray)  
T_mix = T(P, h_mix) ← IF97 Region 2 backward equation
```

The cooling power attributable to the spray is $Q_{\text{spray}} = \dot{m}_{\text{spray}} \cdot (h_{\text{steam}} - h_{\text{mix}})$.

6. Scenario events

Internally every input is a **time series** built from events. The built-in UI exposes
one event time; the engine supports arbitrary lists (the full scenario editor ships in
a coming release):

Event kind	Units	Default behavior
Steam flow	kg/s (UI: t/h)	Step at event time
Spray flow	kg/s (UI: t/h)	Step (or ramp if spray ramp configured)
Spray temperature	K (UI: °C)	Step; must stay in 100–180 °C
Burners firing	count	Ramped over 60 s (configurable), not stepped
Firing fraction	0–1	Step
Steam pressure	Pa (UI: bar)	Step
Steam temperature	K (UI: °C)	Step

The example scenario from the specification is exactly the default scenario set:

t (s)	Event
0	Steam flow = 300 t/h, 2 burners on, spray = 0
300	Steam flow → 450 t/h
400	Spray 0 → 5 t/h
500	Burners 2 → 3

7. Reading the results and charts

Summary numbers (at end of simulation)

Row	Meaning
Outlet T (start → end)	Outlet steam temperature of the uniform coil, start vs end
Metal T (start → end)	Lumped metal temperature (starts at your Metal T input)
Mixed T (end)	Post-spray mixed temperature (equals steam T if no spray)
Velocity (end)	Steam velocity in one tube, m/s
h _i (end)	Internal convection coefficient, W/m ² K
U (end)	Overall heat-transfer coefficient, W/m ² K
Re (end)	Reynolds number in a tube
Q absorbed (end)	Heat picked up by the steam, MW
ρ / cp (end)	IF97 density and heat capacity at the outlet state

Charts — every parameter selectable

After a run, **every thermodynamic parameter** is available as a toggle chip:

Outlet T, Metal T, Mixed T, Steam inlet T, Pressure, Steam flow, Spray flow, Burner

heat, Heat absorbed, Velocity, Re, Pr, Nu, h_i, U, ρ, cp, μ, k, h — tap any combination

to plot them versus time.

How to interpret a step response: after a burner step the metal temperature begins rising immediately, and the outlet follows after a short dead time; after a spray step the outlet begins falling almost immediately (the spray acts at the inlet), settling toward a new steady value.

8. Show calculation — full transparency

Every result must be traceable. Tap **Show calculation details** at the bottom of the results to see, for the end-of-simulation state:

Line	Source
P, T_out	Current steam state at the outlet
ρ	IF97 density at (P, T_out)
cp	IF97 isobaric heat capacity
μ	IF97 dynamic viscosity
k	IF97 thermal conductivity
Pr	IF97 Prandtl number
Re	$\rho \cdot v \cdot D_i / \mu$ from the flow model
Nu	$0.023 \cdot Re^{0.8} \cdot Pr^{0.4}$ (Dittus–Boelter)
$h_i = Nu \cdot k / D_i$	internal convection
U	cylindrical resistance chain

This is a core requirement of the tool (engineering traceability), not a convenience feature.

9. The Validation tab — model comparison

The app ships **three model levels** and runs them side by side on the same scenario:

Model	Description	Purpose
Constant-U	Single block, fixed U (800 W/m²K)	Plant-YAML reference behavior
Dynamic h_i	Single block, h_i from the Re→Pr→Nu chain each step	Isolates the effect of dynamic heat transfer
Uniform (RK4)	Uniform-lumped transient model (metal + steam nodes)	The production model

Tap **Run comparison** to see the end-of-run outlet temperature of all three on the reference scenario (100→150 t/h and 2→3 burners at $t = 300$ s).

The 800 vs ~170 W/m²K discrepancy is surfaced, not hidden. Plant YAML data gives $U \approx 800$ W/m²K at low load, while the Dittus–Boelter chain gives ~170 W/m²K at full load. Both paths are provided as separate, clearly labeled modes — the app never silently blends them. Resolving the gap is a calibration exercise (next section), not something the app decides for you.

10. Calibration against plant data

The **Fit parameters** workflow lets you calibrate two effective parameters against measured plant data:

1. Export a window of historian data containing the **outlet temperature** sampled evenly (the simulation grid is 1 s × 900 s; resample your data to match, or paste a representative series of at least 10 points).

1. Paste the values (°C), comma- or newline-separated, into the text field.
2. Tap **Fit parameters**.

The fitter grid-searches:

- **F_platen** (fraction of furnace release reaching the platen) in 0.05...0.5
- **h_o** (external/furnace-side convection) in 20...500 W/m²K

and reports fit quality:

Metric	Meaning
MAE	Mean absolute error, K
RMSE	Root-mean-square error, K
Max error	Worst instantaneous error, K
Bias	Mean signed error, K (model too hot or too cold overall)
R ²	Fraction of variance explained

Fitted values are labeled empirical and kept separate from the physical parameters — the app never overwrites physics with fitted numbers. Use the fitted F_platen/h_o as the configuration for subsequent simulation runs.

11. The physics inside — equations used

11.1 Property engine (IAPWS-IF97)

All thermophysical properties come from IAPWS-IF97:

- **Region 1** — compressed/subcooled water (feedwater, spray water)
- **Region 2** — superheated steam (main flow, and T(P,h) backward equation)
- **Region 4** — saturation line (Tsat for the spray subcooling check)

Verified in the test suite against the official IAPWS tables, e.g.:

Quantity	State	Official value
h	3 MPa, 300 K	115.331 kJ/kg
h	3 MPa, 500 K	975.542 kJ/kg
h	30 MPa, 700 K	2631.495 kJ/kg
μ	0.1 MPa, 298.15 K	0.890022551 mPa·s
k	0.1 MPa, 298.15 K	0.607509806 W/m·K
Tsat	16.7 MPa	623.988 K

11.2 Flow

$$\begin{aligned}A_{\text{flow}} &= \pi \cdot D_i^2 / 4 & D_i &= 41 \text{ mm} \\ \dot{m}_{\text{tube}} &= \dot{m}_{\text{total}} / (43 \cdot 4) \\ v &= \dot{m}_{\text{tube}} / (\rho \cdot A_{\text{flow}}) \\ \text{Re} &= \rho \cdot v \cdot D_i / \mu\end{aligned}$$

11.3 Heat transfer

$$\begin{aligned}\text{Pr} &= c_p \cdot \mu / k \\ \text{Nu} &= 0.023 \cdot \text{Re}^{0.8} \cdot \text{Pr}^{0.4} & (\text{heating; } n = 0.3 \text{ for cooling}) \\ h_i &= \text{Nu} \cdot k / D_i\end{aligned}$$

Dittus–Boelter validity (Pr 0.7–160, Re > 10 000) is checked and warned, never silently applied outside its range.

11.4 Overall U (cylindrical, per outer area)

$$1/(U \cdot A_o) = 1/(h_i \cdot A_i) + \ln(D_o/D_i)/(2\pi \cdot k_{\text{wall}} \cdot L) + R_{\text{fouling}} + 1/(h_o \cdot A_o)$$

All four resistances are exposed; the dominant one is identified in the results.

11.5 Furnace heat input

$$Q_{\text{platen}} = n_{\text{burners}} \cdot 40 \text{ MW} \cdot \text{firing_fraction} \cdot F_{\text{platen}}$$

F_{platen} default 0.15. Burner changes ramp over 60 s.

11.6 Uniform energy balances

$$\begin{aligned} C_{\text{metal}} \cdot dT_{\text{m}}/dt &= Q_{\text{platen}} - h_{\text{i}} \cdot A_{\text{i}} \cdot (T_{\text{m}} - T_{\text{s}}) - Q_{\text{loss}} \\ C_{\text{steam}} \cdot dT_{\text{s}}/dt &= \dot{m} \cdot c_p \cdot (T_{\text{in}} - T_{\text{s}}) + h_{\text{i}} \cdot A_{\text{i}} \cdot (T_{\text{m}} - T_{\text{s}}) \end{aligned}$$

One metal node and one steam node for the whole coil (uniform model). The steam balance is integrated in exponential-relaxation form with

$\tau = \max(C_{\text{steam}}/(\dot{m} \cdot c_p + h_{\text{i}} \cdot A_{\text{i}}), \Delta t)$ — the same steady state, stable at any timestep.

11.7 Solver

RK4, default $\Delta t = 1$ s (configurable 0.1–2 s). Verified by the timestep-halving convergence test: $dt = 2$ s and $dt = 1$ s agree within 5 K on the reference scenario.

12. Units

Context	Units
UI display	bar, °C, t/h, MW, W/m ² K
Engine internals	SI only: Pa, K, kg/s, J/kg, W, m, kg/m ³ , J/kg·K, Pa·s, W/m·K

Conversions happen exactly once, at the UI boundary. Internally nothing is ever mixed.

13. Error messages and what they mean

Message	Cause	What to do
Spray temperature X °C outside the 100-180 °C plant window	Spray T outside 100–180 °C	Set spray temperature inside the window
Spray water must be subcooled: T=... >= Tsat=... at ... bar	Spray T ≥ Tsat at mixing pressure	Lower spray temperature below Tsat
IF97 state out of range: ...	Pressure/temperature outside IF97 validity (e.g. > 100 MPa, < 0 °C)	Bring inputs into the physical range
Steam flow must be positive	Zero/negative flow entered	Enter a positive flow
Simulation error at t=...	A state went non-physical mid-run	Reduce event magnitude, check initial state

All errors are typed results — the app never crashes on bad input; it explains.

14. FAQ and troubleshooting

Q: The outlet temperature keeps rising the whole run — is that wrong?

A: Probably not. The metal starts at your Metal T input (default 450 °C); if the burner heat input exceeds what the flow carries away, the system is heating toward a higher equilibrium and 15 minutes is not enough to settle. Extend the duration or compare spray-on vs spray-off runs.

Q: Why is tube velocity ~24 m/s at full load?

A: Superheated steam at 167 bar/520 °C has density $\approx 52 \text{ kg/m}^3$ — a tenth of water. With the 41 mm ID this gives $\approx 24 \text{ m/s}$ and $\text{Re} \approx 1.5 \cdot 10^6$ — comfortably turbulent and physically correct.

Q: Why does the metal respond before the outlet?

A: The furnace heats the metal directly; the steam only receives heat through the metal. That cascade delay is the dominant transient behavior of a superheater.

Q: Can I set spray to 200 °C to test the limit?

A: The engine will refuse — that is the plant rule working as intended. Spray water is always subcooled and within 100–180 °C in this plant.

Q: Can I change the metal temperature?

A: Yes — Metal T °C is a user input (default 450 °C) and enters the calculation as the initial temperature of the metal node.

Q: Can I run just one scenario?

A: Yes — every scenario card has its own switch. One switch on = single-scenario run; several switches on = mixed-scenario run. Disabled scenarios never fire.

Q: What pressure should I enter — absolute or gauge?

A: Absolute (bar a). IF97 works in absolute pressure.

Q: Where does the IF97 library come from?

A: `com.hummeling.if97` v2.1.0 by Hummeling Engineering BV, vendored in source form under the LGPL (license file included in the source tree).

15. Building the app from source

```
# Requirements: JDK 17, Android SDK (see local.properties), internet for Gradle deps
git clone https://github.com/mohsenRahimi1708/SteamMixerCalculator.git
cd SteamMixerCalculator # checkout unit4-platen-transient
# the app lives in the platen-superheater/ folder of that branch
./gradlew :app:testDebugUnitTest # 23 tests, all green
./gradlew :app:assembleDebug      # APK at app/build/outputs/apk/debug/app-debug.apk
```

Architecture summary:

```
UI (Compose) → Screens → PlatenSimulator → SteamProperties (IF97)
                    ↓
                FlowModel / HeatTransfer / OverallU
                    ↓
            BurnerModel / SprayModel / ScenarioState
```

The engine package is pure Kotlin/JVM (no Android dependencies) and fully unit-tested; the property engine sits behind an interface so an alternative implementation (e.g. CoolProp) can be swapped in without touching the model.

Document version 2.0.0 — reflects the refactored uniform model (OD 57 / ID 41 mm, 100–180 °C spray window, per-scenario enable/disable, user metal temperature). For the full engineering derivation of every equation, see `ENGINEERING.md` in the repository.